Aditya Gulati

gulati.adi@northeastern.edu LinkedIn GitHub Portfolio

Computer Vision & AI Skills

Languages: Python, C++, SQL
Computer Vision: OpenCV, Object Detection, Image Classification, Multi-Object Tracking, Visual Localization, Camera Calibration
ML Tools: Scikit-learn, Hugging Face Transformers, NumPy, Pandas

Awards

- Won the **Twelve Labs Computer Vision Hackathon** (Apr. 2026) and the **Cursor Hackathon** (May 2026) for Apex AI.
- Won the **Hack@URI AI Track** (Feb. 2026) and the **Yconic Hackathon** (Mar. 2026) for WayfinderAI.

Experience

NU Launch Lab – WayfinderAI (GitHub) Jan. 2026 – Present
Co-Lead, Computer Vision Engineer *Boston, MA*

- Developed a hardware-free indoor navigation system that converts floor plans into connected navigation graphs, eliminating BLE beacon infrastructure and associated installation, calibration, and maintenance costs.
- Built a computer-vision pipeline that extracts room and spatial information from floor plans to generate turn-by-turn shelf directions for visually impaired shoppers, enabling indoor navigation without GPS or physical positioning infrastructure.
- Automated floor-plan labeling and model training, achieving **100% graph connectivity** and a **0.746 direction-accuracy score**, while eliminating manual annotation from the training workflow.

Northeastern Satellite Lab – CubeSat Star Tracker (GitHub) Jan. 2026 – Present
Software Engineer – Computer Vision *Boston, MA*

- Replaced a mathematical star-detection baseline used on small satellites with a CNN-based model, improving detection accuracy on real sky imagery.
- Benchmarked multiple U-Net-based architectures for star detection, selecting ELUNet as the best performer with **91% detection accuracy**, then integrated it with star mapping and catalog matching for CubeSat attitude estimation.
- Solved the model's high compute demand for onboard hardware by designing an INT8-quantized version, keeping performance within **0.3%** of the full model.

Climate Resilient Communities Apr. 2024 – Sept. 2025
GIS Analyst *Toronto, ON*

- Engineered GIS tools for heat-index and flood-risk analysis, reducing local governments' manual climate-risk review time by an estimated **40%**.
- Built an emergency-access mapping tool evaluating where Ontario could add new health clinics, based on 1-hour commute coverage from emergency shelters.

Openlane Oct. 2022 – Mar. 2024
Data Analyst *Toronto, ON*

- Raised data quality from **80% to 95%** across 100,000+ vehicle ownership records and standardized validation across 10+ client accounts, supporting **30% business growth**.

Computer Vision & Robotics Projects

Apex AI (GitHub) | *YOLOv8n, Florence-2, Twelve Labs* Apr. 2026

- Built an AI race engineer with Twelve Labs Pegasus and Marengo that analyzes onboard lap videos, with no sensor telemetry needed, flagging racing line, braking points, and other coaching insights, cutting lap times by an average of **0.75 seconds**.
- Rebuilt the tool to run fully offline and locally, combining motion detection, lightweight vision models, and local language models to flag key moments and generate coaching notes with no cloud dependency.

Edge AI Multisensor System (GitHub) | *Sensor Fusion, Edge Deployment, Ollama* Jun. 2026

- Upgraded an edge hub with a sensor-fusion-based Interval Action Inference framework combining temperature and occupancy data, cutting overheating response time by $\sim$**120**$\times$ in a 30-day simulated run across 26 zones, running locally with no cloud dependency.
- Added an agentic AI layer that learns HVAC scheduling from daily temperature and occupancy data, automating adjustments and reducing manual scheduling effort.

Intelligent Follow-Me Robot (GitHub) | *YOLO26n, MobileNetV2, KCF, Kalman* Jun. 2026

- Developed a person-following robot using camera-based object detection, appearance matching, and visual tracking, maintaining target identity through brief occlusions.
- Implemented Kalman-filter-based tracking and recovery logic to reacquire lost targets, with obstacle avoidance running on low-cost hardware.

Education

Northeastern University Boston, MA, USA
Master's in Artificial Intelligence, GPA: 3.4/4.0 *Expected May 2028*

University of Toronto Toronto, ON, Canada
Honors Bachelor of Science in Statistics and Mathematics *June 2022*